+91 928 450 2604
divyeshunadkat.github.io
divyeshunadkat

# Divyesh Unadkat

## Education

*Ph.D.* — **Computer Science & Engineering**, *Indian Institute of Technology Bombay*, Mumbai. **2023**
CPI: 9.48/10

*B.E.* — **Computer Engineering**, *Dharmsinh Desai University*, Nadiad. **2006 – 2010**
Aggregate: 80.12 %

## Ph.D. Thesis

**Title**: *Techniques for Precise and Scalable Verification of Array Programs*

**Supervisors**: Prof. Supratik Chakraborty , Prof. Ashutosh Kumar Gupta

**Institution**: Indian Institute of Technology Bombay, Mumbai

**Area**: Formal Methods and Software Verification

## Experience

**Principal Engineer, R&D**, *Synopsys*, Hyderabad. **May'26–Present**

**Senior Staff Engineer, R&D**, *Synopsys*, Hyderabad. **Feb'24–Apr'26**

**Staff Engineer, R&D**, *Synopsys*, Hyderabad. **Aug'23–Jan'24**

**Scientist/Senior Software Engineer**, *TCS Research*, Pune. **Jun'21–Jul'23**

**Researcher/Software Engineer**, *TCS Research*, Pune. **Jun'10–May'21**

**Research Intern**, *TCS Research*, Pune. **Dec'09–Apr'10**

## Technical Skills

**Programming**: C/C++, Python, Java

**Compilers**: LLVM, Clang, GNU Tool Chain (GCC, GDB, Make)

**Research Tools**: Z3, CVC5, CBMC, Daikon, CPAChecker, InvGen

**Development Tools**: Cursor, VS Code, Emacs, Vim

**Version Control**: Git, Perforce, CVS

**Telecommunication Tools**: Microsoft Teams, Zoom, Google Meet, Slack

## Tool Dev

*Diffy* — Generalized Inductive Reasoning for Arrays. Published in CAV 2021 [3]. *repository*

*Vajra* — Full-Program Induction. Published in TACAS 2020 [4, 5], STTT 2022 [2]. *repository*

*Tiler* — Verifying Array Programs by Tiling. Published in SAS 2017 [6]. repository

*DIV* — Dynamic Inference Verifier. Internal Tool, TCS Research. Published in HVC 2013 [8]

*ScaleM* — Scaling Model Checking with Abstractions Inferred using Dynamic Analysis. Internal Tool, TCS Research. Published in ICST 2013 [7]

*AutoGen* — Automatic Test-case Generation using Model Checking. Internal Tool, TCS Research

## Publications


[1] Divyesh Unadkat. Techniques for Precise and Scalable Verification of Array Programs. *Doctoral Dissertation, IIT Bombay*, December 2022.

[2] Supartik Chakraborty, Ashutosh Gupta, and Divyesh Unadkat. Full-Program Induction: Verifying Array Programs sans Loop Invariants. In *International Journal on Software Tools for Technology Transfer (STTT)*, pages 843–888, September 2022.

[3] Supartik Chakraborty, Ashutosh Gupta, and Divyesh Unadkat. Diffy: Inductive Reasoning of Array Programs using Difference Invariants. In *Proc. of the 33rd International Conference on Computer-Aided Verification (CAV)*, pages 911–935, 2021.

[4] Supartik Chakraborty, Ashutosh Gupta, and Divyesh Unadkat. Verifying Array Manipulating Programs with Full-Program Induction. In *Proc. of the 26th International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, pages 22–39, 2020.

[5] Mohammad Afzal et. al. VeriAbs : Verification by Abstraction and Test Generation (Competition Contribution). In *Proc. of the 26th International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, pages 383–387, 2020.

[6] Supratik Chakraborty, Ashutosh Gupta, and Divyesh Unadkat. Verifying Array Manipulating Programs by Tiling. In *Proc. of the 24th International Static Analysis Symposium (SAS)*, pages 428–449, 2017.

[7] Anand Yeolekar et. al. Scaling Model Checking for Test Generation using Dynamic Inference. In *Proc. of the 6th International Conference on Software Testing, Verification and Validation (ICST)*, pages 184–191, 2013.

[8] Anand Yeolekar and Divyesh Unadkat. Assertion Checking using Dynamic Inference. In *Proc. of the 9th Haifa Verification Conference (HVC)*, pages 199–213, 2013.


## Awards

**Individual Award**: Spot Award

**Institution**: Synopsys, Hyderabad

**Description**: Performance par excellence. FY 2024.

**Individual Award**: Conference Travel Grant

**Institution**: ACM India

**Description**: Selected among only 50 national and international participants for the inaugural edition of ACM Pingla Interactions in Computing 2024 held at Infosys Mysore.

**Team Award (Recurring)**: Best Verification Tool

**Institution**: International Software Verification Competition (SV-COMP)

**Description**: VeriAbs [5] won SV-COMP in 2022, 2021, and 2020. My contribution included the development of array verification tools Diffy [3], Vajra [4, 2] and Tiler [6] which are integrated in VeriAbs. Refer to [5] and [1] for more details.

**Individual Award**: Special Mention on CSE IITB Webpage

**Institution**: Indian Institute of Technology Bombay, Mumbai

**Description**: SV-Comp win acknowledged on CSE IITB news page. Posted on 22-12-2019.

**Individual Award**: Most Admired Sprint Thesis Talk

**Institution**: Indian Institute of Technology Bombay, Mumbai

**Description**: Runner-up, Senior Researcher Sprint Talks, RISC 2017, IIT Bombay.

**Individual Award**: Best Speaker in Sprint Thesis Talk

**Institution**: Indian Institute of Technology Bombay, Mumbai

**Description**: Winner, Early Researcher Sprint Talks, RISC 2016, IIT Bombay.

**Individual Award**: Eklavya Gold Medal

**Institution**: Dharmsinh Desai University, Nadiad

**Description**: Highest aggregate in first four semesters of computer engineering, 2008.

## Projects

*Generalized Full-Program Induction*

We devised an inductive reasoning technique that further generalizes the *full-program induction* technique. Significantly, we simplify the inductive step of the analysis. The method can prove a sub-class of quantified as well as quantifier-free assertions in array programs with sequentially composed as well as nested loops. Like full-program induction, this technique also inducts over the entire program via the program parameter $N$ and infers *difference invariants* between two slightly different versions of a program during the inductive step. We have implemented it in the tool *Diffy* and demonstrated its effectiveness vis-a-vis award winning verification tools on a set of array benchmarks from SV-COMP. Refer publications [1, 3] for details.

Timeline: 2020 – 2021

Language/Tool: C++/LLVM, SMTLIB2/Z3

Team size: One

*Full-Program Induction*

Formally verifying properties of programs that manipulate arrays of parametric size in loops is computationally challenging. In this project, we designed the novel *Full-Program Induction* technique to overcome this challenge. The technique inducts over the entire program via the program parameter $N$ and computes the difference of programs and properties with different parameter values during the inductive step. We have implemented it in the tool *Vajra*. We compare its performance vis-a-vis state-of-the-art verification tools on a set of array manipulating benchmarks. Refer publications [1, 2, 4, 5] for details.

Timeline: 2018 – 2020

Language/Tool: C++/LLVM, SMTLIB2/Z3

Team size: Two

*Verification by Tiling*

We developed a novel property-driven verification method that can infer array access patterns in loops, and use this information to compositionally prove universally quantified assertions about arrays. We have implemented this technique in a tool called *Tiler*, written in C++, which outperforms several state-of-the-art tools on a suite of interesting benchmarks. Refer publication [1, 6] for details.

Timeline: 2016 – 2017

Language/Tool: C++/Clang, Java/Daikon, SMTLIB2/Z3

Team size: One

*Dynamic Inference and Verification*

Model checkers often run into scalability issues when verifying programs with a large state space. To address this challenge, several abstraction-based techniques have been proposed in the literature. We have developed an innovative CEGAR technique that generates refinement predicates using dynamic analysis. Our verification approach, implemented in *DIV*, using the Java programming language, is sound and compositional, even though the underlying dynamic analysis may return unsound predicates. We have demonstrated its effectiveness on challenging benchmarks. Refer publication [8] for details.

Timeline: 2013 – 2014

Language/Tool: Java/Daikon, C++/Kvasir, C++/CBMC

Inhouse Tools: C++/(AutoGen, Misra-CFE), Java/(PRISM, Slicer, Unparser)

Team size: Two

*Scaling Model Checking*

Automatic testcase generation tools seldom scale to large systems. We proposed a new approach that uses source-level abstractions generated using dynamic analysis to abstract parts of code to scale the testcase generation effort based on model checking. Our tool *ScaleM*, written in Java, summarizes complex code fragments to enable the generation of application-level tests in large-size C code using inferred program properties. Refer publication [7] for details.

Timeline: 2012 – 2013

Language/Tool: Java/Daikon, C++/Kvasir, C++/CBMC

Inhouse Tools: C++/(AutoGen, Misra-CFE), Java/(PRISM, Slicer, Unparser)

Team Size: Three

*Automatic Testcase Generation*

Rigorous verification of safety critical systems is inevitable. Manual testing of such systems is inefficient as well as inadequate. We have developed *AutoGen*, written in C++, a fully automatic, structural testcase generation tool for C programs that uses a model checker at the back-end. The tool is capable of generating testcases satisfying various coverage criteria including the modified condition/decision coverage (MC/DC) mandated by ISO 26262 & DO178B standards.

Timeline: 2010 – 2011

Language/Tool: C++/CBMC

Inhouse Tools: C++/Misra-CFE, Java/(PRISM, Slicer, Unparser)

Team size: Five

*Statechart Translation*

Harel statecharts are widely used to model real time reactive systems using the Statemate tool. During my internship, we developed a tool for translating various statecharts into Symbolic Analysis Laboratory (SAL) specification. Statecharts are first converted into an Intermediate Representation (IR). We then unparse the IR to SAL specification. The motive was to enable verification of statecharts for erroneous conditions like Non Determinism, Race Conditions and State Reachability using the relatively more scalable model checker available in the SAL framework, sal-bmc.

Timeline: 2009 – 2010

Tools & Languages: SAL, Statemate, lex, yacc, bison

Inhouse Tool: ACH-SCH Parser

Team size: Two

## Conference Presentations

**Diffy: Verifying Array Programs using Difference Invariants**: 33rd International Conference on Computer Aided Verification (CAV), Los Angeles, USA (*Online*), July 2021

**Verifying Array Manipulating Programs with Full-Program Induction**: 26th International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS), Luxembourg (*Online*), March 2021

**Verifying Array Manipulating Programs by Tiling**: 24th International Static Analysis Symposium, SAS, New York, USA, August 2017

**Assertion Checking using Dynamic Inference**: 9th Haifa Verification Conference, Haifa, Israel, November 2013

## Invited Talks

**Dance of the Dragons: Induction, Difference Computation and SMT Solving**: Formal Methods Update Meeting, IIT Delhi, July 2022

**Difference Invariants for Inductive Verification**: 6th Indian SAT+SMT School (*Online*), December 2021

**Exploiting Induction and Difference Computation to Verify Array Programs**: Formal Methods Update Meeting (*Online*), July 2021

**The Full-Program Induction Technique**: 5th Indian SAT+SMT School, IIT Hyderabad (*Online*), December 2020

**Verifying Array Manipulating Programs with Full-Program Induction**: Software Engineering Research India (SERI), IIIT Hyderabad (*Online*), July 2020

**Lightening Talk: Verifying Array Manipulating Programs by Tiling**: 2nd Indian SAT+SMT School, Infosys Campus, Mysuru, December 2017

## Competition Talks

**Verifying Array Manipulating Programs by Full-Program Induction**: Research and Innovation Symposium in Computing, RISC 2019, IIT Bombay

**Verifying Array Manipulating Programs by Tiling**: Sprint Thesis Talk, Research and Innovation Symposium in Computing, RISC 2017, IIT Bombay

**Towards Precise Software Verification**: Sprint Thesis Talk, Research and Innovation Symposium in Computing, RISC 2016, IIT Bombay

## Poster Presentations

**Full-Program Induction: A paradigm shift in the Verification of Array Programs**: ACM Pingla Interactions in Computing, Infosys, Mysore, February 2024

**Verifying Array Programs with Full-Program Induction**: 4th Indian SAT+SMT School, IIT Bombay, December 2019

**Executive Summary on Tiling to Verify Array Programs** : TCS Anvetion Workshop, IIT Madras Research Park, Chennai, 2018

**Verifying Array Manipulating Programs by Tiling**: Research and Innovation Symposium in Computing, RISC, IIT Bombay, Mumbai, March 2017

## Interests

**Sports**: Table Tennis, Volleyball, Football

**Recreation**: Yoga, Novels, Music, Movies

## Links

**Webpage**: https://divyeshunadkat.github.io/

**LinkedIn**: https://www.linkedin.com/in/divyeshunadkat/

**GitHub**: https://github.com/divyeshunadkat/

**dblp**: https://dblp.uni-trier.de/pers/hd/u/Unadkat:Divyesh

**Scholar**: https://scholar.google.co.in/citations?user=8d48NqMAAAAJ

## Contact

**Mobile**: +91 928 450 2604
**E-Mail**: divyeshunadkat001@gmail.com

## References

Available upon request.